About Saeem Mehmood Sadiq

Quick Facts About Saeem

- **Full name:** Saeem Mehmood Sadiq
- **Role:** Senior Software Engineer (backend), moving into Data and AI Engineering
- **Current company:** ILI Digital
- **Experience:** 5+ years of professional software engineering, starting December 2020
- **Primary languages:** Go, Elixir, Python
- **Location:** Lahore, Pakistan
- **Email:** saimmahmood61@gmail.com
- **LinkedIn:** https://www.linkedin.com/in/saimmahmood61
- **GitHub:** https://www.github.com/saim61
- **Booking a call:** https://calendar.app.google/czDkhWsVGq97Xqa47

Professional Summary of Saeem

Saeem Mehmood Sadiq is a senior backend engineer with over five years of experience building high-performance systems in Go, Elixir, and Python. He has worked across real-time systems, iGaming, ticketing platforms, SaaS products, and distributed architectures. Saeem is known for clean engineering, pragmatic decision-making, and mentoring junior developers, and he typically owns projects end-to-end, from gathering requirements through to production reliability.

Saeem's Current Role — ILI Digital

Period: April, 2026 – Present

Saeem currently works at ILI Digital as a Senior Software Engineer, where he manages projects built in Go, including a Go-based project for automated replies. Alongside this role, he is transitioning into Data and AI Engineering by taking on tasks in his spare time.

Saeem's Experience at Code Graphers (Senior Software Engineer)

Period: June 2024 – March 2026

At Code Graphers, Saeem worked as a Senior Software Engineer, mainly in Go, across software asset management and iGaming projects.

- Re-engineered the Python `imagehash` library in Go, keeping output parity across all hash types. This enabled a latency-critical image-matching pipeline to migrate to a faster Go backend.
- Built backend services in Go for a client's software asset management platform, improving reliability and observability.
- Reduced the client software's idle RAM usage from about 2 GB to about 40 MB (roughly a 98% reduction) by finding memory hotspots and leaks and optimizing allocations and data structures so nothing was retained unnecessarily while idle.
- Built a small serverless pipeline for a client's test task: an AWS Lambda function, scheduled with EventBridge, periodically fetched currency data from an open-source source and stored it in DynamoDB, deployed with CloudFormation.

- Led feature development, bug resolution, and operational oversight for 13 microservices in an iGaming ecosystem.
- Wrote tests across those 13 microservices, raising test coverage from about 10% to about 85%.
- Worked with senior engineers and external provider teams to architect and integrate two new live-casino provider integrations.
- Mentored junior developers on Go best practices and code review standards.

How Saeem Reduced Idle RAM from 2 GB to 40 MB at Code Graphers

At Code Graphers, Saeem worked on the Go codebase of a client's agent software, which used about 2 GB of RAM even while idle. He first studied the codebase to understand how the whole system worked, then profiled the agent with pprof, both while it was working and after it went idle.

The profiles pointed to three main problems:

- Long-running infinite loops that kept running after the agent's work was done.
- The same regular expressions being compiled more than 100,000 times. Saeem made each one compile once and be reused.
- API calls that fetched more than 150,000 records at once without pagination. He added pagination.

Fixing these meant reworking a large part of the codebase, all in Go. To measure each change, he built the agent, let it finish its work on a machine, waited another 30 to 40 minutes, and then checked its idle RAM. Idle memory dropped from about 2 GB to about 40 MB, a reduction of roughly 98%.

How Saeem Ported the imagehash Library from Python to Go

At Code Graphers, a client asked Saeem to rewrite the Python imagehash library in Go. He mirrored the Python implementation closely and rewrote parts of the math himself, because Go's math libraries gave results that differed from Python's by a bit or two. To prove the two versions matched, he ran the same sample image through every hashing algorithm in both languages and compared the outputs side by side. The Go version ran significantly faster. While comparing outputs, he found a case where his Go version was correct but the Python library had a bug in its ColorHash algorithm. He fixed it upstream, and his pull request #216 was accepted: https://github.com/JohannesBuchner/imagehash/pull/216

Saeem's Work on 13 iGaming Microservices at Code Graphers

On an iGaming platform of 13 Go microservices, Saeem's day-to-day work was fixing bugs, reviewing pull requests, building new features, and working closely with senior engineers. One of his biggest and most time-consuming tasks was writing tests for the Go code across all 13 services, which raised test coverage from about 10% to about 85%. While building two live-casino provider integrations, the hardest part was understanding how Kafka was used: it connected all 13 services, which made events hard to trace and bugs hard to debug.

Saeem's Experience at Dycoders (Senior Software Engineer, Hybrid)

Period: July 2023 – January 2024

At Dycoders, Saeem led an Elixir/Phoenix engineering team.

- Led the team through sprints, architecture discussions, and CI/CD workflows.

- Established testing standards and code review processes, which reduced production bugs and improved deployment reliability.
- Built and optimized Phoenix and LiveView backends with a focus on maintainability and real-time performance.
- Mentored junior engineers through pair programming and structured feedback.

The team had about 10 backend and frontend engineers, led by Saeem together with a colleague. The two of them managed the team and also took on the hardest programming tasks. Every module had to come with its own tests, and CI/CD ran those tests before code could be merged. QA engineers added manual and automated testing on top.

Saeem's Experience at Automatiq (Senior Software Engineer, Contract, Remote)

Period: May 2022 – June 2023

At Automatiq, Saeem focused on integrating external providers into a ticketing platform.

- Onboarded more than 10 new providers by integrating external systems end-to-end, covering APIs, workflows, and edge cases, which expanded platform coverage and supported business growth.
- Delivered integrations with high operational stability through resilient error handling, retries and timeouts, and clear observability for faster production debugging.
- Improved response times and platform performance by optimizing backend endpoints and database interactions so the platform handled peak usage reliably.

The platform received ticket emails from many ticketing providers. Saeem's team analysed each provider's email format and extracted details such as the event name, venue, seat numbers, and number of people, which were saved in the database as a memo. The hardest were two providers whose email formats kept changing, so the team built ways to detect which format an email used and extract the details accordingly.

Saeem's Experience at InvoZone (Software Engineer, Hybrid)

Period: December 2020 – May 2022

InvoZone was where Saeem started his professional career, working on real-time gaming systems.

- Developed Elixir and Go microservices for real-time gaming systems using optimized concurrency patterns.
- Implemented gRPC and GraphQL services to improve modularity and scalability.
- Owned backend logic for multiple game modules and mentored junior engineers.

Saeem's High Score Games Module at InvoZone

At InvoZone, Saeem worked in Elixir on a real-time competition app where two users competed and the winner received a prize. He was solely responsible for its high score games module: event-based and time-based games where users scored points and the top players won prizes. An admin panel controlled the number of winners, the start and end times, when prizes were distributed, and how. Saeem built all of it, from background jobs (using Oban) to real-time leaderboards to prize distribution. Running many scheduled jobs concurrently and in parallel with very few issues showed him the strength of Elixir.

Saeem's Career Timeline

Saeem's jobs in order, from the most recent to his first job:

1. ILI Digital: Senior Software Engineer, April 2026 – Present
2. Code Graphers: Senior Software Engineer, June 2024 – March 2026
3. Dycoders: Senior Software Engineer (Hybrid), July 2023 – January 2024
4. Automatiq: Senior Software Engineer (Contract, Remote), May 2022 – June 2023
5. InvoZone: Software Engineer (Hybrid), December 2020 – May 2022

Saeem's Technical Skills

- **Backend languages:** Go, Elixir (Phoenix, OTP, LiveView), Python (FastAPI, Django)
- **Architecture:** Microservices, event-driven systems, REST, gRPC, GraphQL, WebSockets
- **Databases and data stores:** PostgreSQL, MySQL, DynamoDB, Redis
- **Data engineering:** Batch ETL pipelines, dbt, PySpark, Parquet, star-schema modeling, slowly changing dimensions (SCD2), data quality checks, idempotent and incremental loading with watermarks
- **Cloud:** AWS Lambda, EventBridge, DynamoDB, CloudFormation
- **Infrastructure and DevOps:** Docker, Docker Compose, Kubernetes, Helm, GitHub Actions, CI/CD
- **Observability:** Structured logging, Prometheus metrics, OpenTelemetry
- **Engineering practices:** Testing and TDD, code review, performance and memory optimization
- **Domains:** Real-time systems, iGaming, ticketing, SaaS, software asset management

Saeem's Personal Projects and Open Source Work

All of Saeem's personal projects are public on his GitHub: https://www.github.com/saim61

Saeem's projects are Hookline (a webhook delivery service in Python), Podium (a real-time leaderboard system in Go), fxflow (a foreign exchange rates ETL pipeline), the Olist Analytics Platform (an e-commerce data warehouse), and a Go messaging app. His open-source contribution is a bug fix in the ColorHash algorithm of the Python imagehash library (pull request #216, accepted).

Hookline — Webhook Delivery Service (Python)

Hookline is Saeem's reliable webhook delivery service, a smaller, readable implementation of what products like Svix and Hookdeck offer. It accepts an event in milliseconds and then delivers it to customers' HTTP endpoints with retries, exponential backoff with jitter, per-endpoint circuit breakers, HMAC signatures, idempotency keys, a dead letter queue, and a permanent audit trail of every delivery attempt. The delivery worker uses a transactional outbox drained with `SELECT ... FOR UPDATE SKIP LOCKED`, so multiple workers can run concurrently without double-delivering. It includes an authenticated API, an operator dashboard, Prometheus metrics, and a Helm chart, and is covered by 292 tests at 94% coverage. Saeem built it for practice and to broaden his expertise.

- **Stack:** Python, FastAPI, PostgreSQL, SQLAlchemy (async), Redis, Alembic, HTMX, structlog, Prometheus, OpenTelemetry, Docker, Helm, GitHub Actions
- **Link:** https://github.com/saim61/hookline

Podium — Real-Time Leaderboard System (Go)

Podium is Saeem's real-time leaderboard system written in Go. Five games run on the server as state machines, so the server computes every score itself and clients can never submit a forged score. Scores are stored durably in PostgreSQL, while Redis sorted sets act as a derived read model that can be rebuilt from Postgres at any time. Leaderboard updates reach browsers over WebSockets and travel through Redis Pub/Sub, so real-time updates keep working across multiple server instances. It also includes Argon2id

authentication with refresh token rotation, period reports, rate limiting, Prometheus metrics, an OpenAPI description, and a playable demo page, and is covered by 429 tests. Podium started as a backend challenge Saeem found online; he built it for practice and to broaden his expertise.

- **Stack:** Go, PostgreSQL, Redis, WebSockets, sqlc, Docker, Prometheus
- **Link:** https://github.com/saim61/podium

fxflow — Foreign Exchange Rates ETL Pipeline (Python)

fxflow is Saeem's batch ETL pipeline for foreign exchange rates, built to learn data engineering fundamentals from first principles before using standard tooling. It pulls daily European Central Bank reference rates from the Frankfurter API, stores them in an immutable, append-only raw layer, transforms them into an analytics-ready fact table with dbt, validates them with data quality tests, and serves them through a FastAPI REST API. The pipeline is idempotent, uses watermarks to fetch only missing data and backfill gaps, and records every run for observability.

- **Stack:** Python, PostgreSQL, SQLAlchemy, dbt, FastAPI, Docker
- **Link:** https://github.com/saim61/fxflow

Olist Analytics Platform — E-Commerce Data Warehouse (Python)

The Olist project is Saeem's end-to-end batch data pipeline built on the public Brazilian e-commerce dataset by Olist. It turns raw CSV data into a tested star-schema warehouse and business-ready Parquet marts, answering questions such as which product categories and states drive revenue, how revenue is trending, and where deliveries are late. Data moves through source, raw, staging, and warehouse layers, with incremental extraction using watermarks, a customer dimension tracked as a slowly changing dimension (SCD2), quality gates that abort the run if checks fail, and PySpark jobs that build partitioned Parquet marts. The whole platform is reproducible with Docker.

- **Stack:** Python, PostgreSQL, PySpark, Parquet, pandas, Docker
- **Link:** https://github.com/saim61/olist

Go Message App

A real-time messaging backend built by Saeem with Go, WebSockets, Clean Architecture, JWT authentication, and PostgreSQL.

imagehash — Open-Source Contribution

Saeem fixed a bug in the ColorHash algorithm of the Python `imagehash` library. His pull request #216 was accepted.

Saeem's University Education

Saeem holds a BSc in Computer Science from FAST-NUCES (National University of Computer and Emerging Sciences), Islamabad campus.

His final year project was a Twitter Cyber Bullying Detector, which analyzed tweets from a user's Twitter home feed and alerted users about accounts that were potential bullies. During university, Saeem particularly enjoyed the Software for Mobile Devices course, where he learned to build native Android apps, as well as mathematics courses.

Saeem's A Levels

Saeem completed his A Levels in 2013 at Arab Unity School in Dubai, UAE, under the Edexcel board. His subjects were Mathematics, Physics, and Chemistry.

Saeem's O Levels

Saeem completed his O Levels in 2011 at The Westminster School in Dubai, UAE, under the Cambridge (CIE) board. His subjects were Mathematics, Physics, Chemistry, Information and Communication Technology, Economics, English, and Urdu.

Saeem's Background

Saeem was born and raised in the United Arab Emirates, where he completed both his O Levels and A Levels in Dubai. He then studied Computer Science at FAST-NUCES in Islamabad, Pakistan, and is now based in Lahore, Pakistan.

What Saeem Is Learning and Where His Career Is Heading

Alongside backend engineering, Saeem is transitioning into Data and AI Engineering using Python. He is also interested in Platform Engineering, Security Engineering, and AI Systems Engineering as future directions. His fxflow and Olist projects are part of this data engineering journey, and this "About Saeem" assistant is itself one of his learning projects, built to understand embeddings and retrieval-augmented generation (RAG) hands-on.

Saeem's Hobbies and Fun Facts

Outside of work, Saeem enjoys going out with friends and loves hiking. He is especially drawn to the mountains and enjoys a good hike that ends with a killer view. He has travelled to many places in the Northern Areas of Pakistan and has also explored the hilly areas around Baku, Azerbaijan. He also likes to hang out with his friends in his free time or watch movies/seasons

His most memorable trips were two university trips to Hunza and Gilgit, where his whole batch travelled together. On both trips, everyone sat around a bonfire under the stars and sang songs for hours.

How Saeem Approaches a New Codebase or a Hard Bug

When Saeem joins a new codebase, he first uses AI tools such as Claude to understand the project and get a quick picture of its architecture. He then runs the code locally and traces how it flows, adding print statements and reading the log files. Once he understands the problem, he writes a plan for the fix and works through it, always checking for ripple effects on the rest of the system.

Saeem's Favourite Programming Languages: Go and Elixir

Saeem enjoys both Go and Elixir, but if he had to pick one, he would choose Go, because it is both powerful and simple. He finds Elixir equally powerful, but its functional style makes it harder than more conventional languages.

How to Contact Saeem

The best way to reach Saeem is by email at saimmahmood61@gmail.com. He can also be found on LinkedIn and GitHub, and anyone can book a meeting with him through his calendar.

- **Email:** saimmahmood61@gmail.com
- **LinkedIn:** https://www.linkedin.com/in/saimmahmood61
- **GitHub:** https://www.github.com/saim61
- **Booking a call:** https://calendar.app.google/czDkhWsVGq97Xqa47